Université Paris Dauphine – PSL

M2 MATH

Regularity structures for the singular stochastic partial differential equation ϕ^4_3

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Abstract

This master's thesis explores the mathematical foundations of singular stochastic partial differential equations (SPDEs), focusing on the stochastic quantization of the classical ϕ^4_3 model in quantum field theory. Due to the extremely rough nature of the driving space-time white noise, traditional calculus fails because the non-linear term ϕ^3 is mathematically ill-defined. To overcome these singularities, we investigate Martin Hairer's landmark theory of regularity structures. Specifically, we study a recent, cutting-edge "diagram-free" approach that replaces complex combinatorial trees with a more parsimonious index set of multi-indices. By treating the directional Malliavin derivative as a "tangent vector" to the solution manifold, it can be mathematically analyzed as a modeled distribution. This geometric reformulation, in conjunction with a spectral gap inequality, yields robust stochastic estimates and BPHZ renormalization. Our work provides an overview of the theory of regularity strucutures, and detailed reformulation of certain proofs of this modern approach.

Contents

Chapter 1

Introduction

Let us define the parabolic differential operator of second order :

$$L \coloneqq \partial_0 - \sum_{i=1}^{d} \partial_i^2.$$

Here, x_0 is the time variable and $\{x_i\}_{i=1,\dots,d}$ are the space variable. We are insterested in the equation

$$L\phi - \lambda\phi^3 = \xi \quad \text{on } \mathbb{R}^d \tag{1.1}$$

where λ is a real parameter, and ξ a space time function/distribution. This equation presents a non-linearity (when $\lambda \neq 0$). We are insterested in the cases where ξ is rough, and thus the equation does not have a rigourous meaning a priori. Indeed, if ξ is a distribution so rough that the solution is also not better that a distribution, then ϕ^3 is ill-defined. For example, if $\lambda = 0$, $d = 0$ and $\xi = \frac{d^2B}{dx_0^2}$, then the solution Π_0 is given by $\frac{dB}{dx_0}$ which is a genuine distribution of regularity just worse than $-\frac{1}{2}$ (Brownian motion is $\frac{1}{2}^-$-Hölder and derivation lowers the regularity by one). Now, multiplying two distributions of respective regularity β, β' only makes canonical sense when $\beta + \beta' > 0$, this is Young's multiplication theorem.

Now in the case we are interested in, where we have at least a space variable $d \geq 1$, and we consider space-time white noise (to be defined rigourously later). We will define Hölder-Besov spaces precisely later, but heuristicly : a distribution is of regularity β if, under the parabolic rescaling $(x_0, x_i) \mapsto (\epsilon^2 x_0, \epsilon x_i)$, it gets multiplied by ϵ^β.

ξ has regularity $-\frac{D}{2}^-$, so the higher the dimension, the rougher it is.

Since L is a parabolic operator of order 2, its inverse L^{-1} gains two levels of regularity (we will prove this precisely later, via Schauder estimates on the heat kernel). Hence the linear solution $\Pi_0 \coloneqq L^{-1}\xi$ has regularity

$$\alpha \coloneqq 2 - \frac{D}{2} = \frac{2-d}{2}.$$

Expecting ϕ to be no better than Π_0, we need $2\alpha > 0$ to even give a meaning to ϕ^2, hence to ϕ^3.

But $\alpha = \frac{2-d}{2} \leq 0$ as soon as $d \geq 2$: in this regime, the non-linear term ϕ^3 has a priori no meaning, and (1.1) is purely formal.

A first step towards solving this problem is using renormalization. To understand this, we will discuss the Da Prato-Debussche approach in a simpler setting. Consider the equation

$$(-\Delta)\phi = \lambda\phi^3 + \xi, \quad \text{on } B \subset \mathbb{R}^4. \tag{1.2}$$

We replaced the heat operator by the Laplacian, because its ellipticity simplifies the discussion. We choose $d = 4$ so that ξ has regularity -2^-, thus we expect ϕ to have regularity O^- so that we are in the singular setting where ϕ^3 does not make sense a priori.

The idea is to write $\phi = \Pi_0 + \psi$ where Π_0 solves the linear equation $(-\Delta)\Pi_0 = \xi$, and hope that because - at least for small λ - Π_0 approximates well ϕ, ψ would have better regularity that ϕ. We can write the equation satisfied by ψ :

$$(-\Delta)\psi = \lambda\phi^3 = \lambda(\psi^3 + 3\psi^2\Pi_0 + 3\psi\Pi_0^2 + \Pi_0^3)$$

Now if ψ is regular enough so that we can make sense of its powers, this is an improvement. Indeed, before we had to make sense of ϕ^3 for a general ϕ of negative regularity, and now it suffices to make sense of Π_0^2 and Π_0^3 where Π_0 is an explicit distribution defined by $(-\Delta)\Pi_0 = \xi$. Of course, Π_0 is still of regularity 0^-, so Π_0^2, Π_0^3 are still analytically ill-defined, but we expect to be able to construct thanks to probabilistic arguments. Suppose we succeed, and Π_0^i has regularity $\alpha \times i$ for $i = 1, 2, 3$. Then we expect ψ to have regularity $3\alpha + 2$. If $5\alpha + 2 > 0$ (and as a consequence $2(3\alpha + 2) + \alpha > 0$), Young's multiplication theorem allows us to make sense of the products $\psi\Pi_0^2$ and $\psi^2\Pi_0$. We will thus be able to set up a fixed point argument to find ψ and as a consequence $\phi = \psi + \Pi_0$. To sum up, the approach is comprised of a probabilistic step, where we need to construct new objects, that are iterated integrals of the noise, and an analytical step where we built the solution. The more involved approach of regularity structures follows the same scheme.

The probabilistic construction of Π_0^2 and Π_0^3 is based on renormalization. First, regularise the noise by convolution against a regularising kernel η : $\xi_\rho := \xi \star \eta_\rho$. Then, denote $\Pi_{0,\rho}$ the solution for the regularised equation $(-\Delta)u = \xi_\rho$. This is a smooth function and we can take its powers $\Pi_{0,\rho}^i$, $i = 2, 3$.

It turns out $\mathbb{E}\left[\Pi_{0,\rho}^2(x)\right] = C_\rho$ independent of x, and C_ρ diverges when $\rho \to 0$, so these powers do not converge when we take out the regularisation. But the renormalized $\Pi_{0,\rho}^{\diamond 2} := \Pi_{0,\rho}^2 - C_\rho$ do have a variance uniformly bounded in ρ, in fact, if actually converges to a random distribution $\Pi_0^{\diamond 2}$ when $\rho \to 0$. Similarly, $\Pi_{0,\rho}^{\diamond} := \Pi_{0,\rho}$ and $\Pi_{0,\rho}^{\diamond 3} := \Pi_{0,\rho}^3 - 3C_\rho\Pi_{0,\rho}$ have uniformly bounded variance in ρ, and converge to some distributions Π_0 and $\Pi_0^{\diamond 3}$. And these limit most importantly do not depend on the choice of regularisation η_ρ, which legitimates those renormalization.

The Da Prato-Debussche approach on the original equation, with $d = 3$ does not work, because we do not get an equation for ψ where Young's multiplication theorem applies, assuming we can construct the iterated integrales. Trying to further decompose $\psi = Z + \theta$ is of no help, because we run in the same issue when writing the equation for θ. We won't be able to find an equation where we can perform a fixed point argument. This is the starting point of the theory of regularity structures introduced by Martin Hairer in [4].

Chapter 2

Useful tools

2.1 Distributions and Hölder regularity

As we saw in our heuristic regularity power counting, we will have to work with *functions* of negative regularity, i.e. distributions. Consider the space $\mathscr{D}(\mathbb{R}^d) \coloneqq C_c^\infty(\mathbb{R}^d)$ of smooth functions with compact support on $\mathbb{R}^d$. The space of distribution $\mathscr{D}'(\mathbb{R}^d)$ is the dual of $\mathscr{D}(\mathbb{R}^d)$: a distribution T on $\mathbb{R}^d$ is a linear form $\mathscr{D}(\mathbb{R}^d) \longrightarrow \mathbb{R}$, such that for all compact $K \subset \mathbb{R}^d$, there exists $r_k \in \mathbb{N}$ with

$$|T(\phi)| \lesssim ||\phi||_{C^r} \coloneqq \max_{|k|\leq r} ||D^k\phi||_\infty \quad \forall \phi \in \mathscr{D}$$

We saw that distributions generalise function because every locally integrable function f defines a distribution by integration :

$$\forall \phi \in \mathscr{D}, \quad f(\phi) \coloneqq \int_{\mathbb{R}^d} f(x)\phi(x)\,\mathrm{d}x.$$

Of course $\mathscr{D}'$ is a vector space. We can also define derivatives of arbitrary order $k = (k_1, \dots, k_d)$ of a distribution T by setting :

$$\forall \phi \in \mathscr{D}, \quad D^k T(\phi) \coloneqq (-1)^{|k|} T(D^k\phi).$$

when T is a differential function, this corresponds to integration by part.

For $\phi \in C^\infty$ and $T \in \mathscr{D}'$, we can define the product $\phi \cdot T$ by :

$$\forall \psi \in \mathscr{D}, \quad (\phi \cdot T)(\psi) \coloneqq T(\phi\psi) \quad (\text{indeed, } \phi\psi \in \mathscr{D}).$$

But in general, there is no canonical way of multiplying two distriubtions. This is really the heart of the problem of singular SPDEs. To solve this, we will need to formalise a notion of regularity for distributions.

Remarque 2.1. Let $r \in \mathbb{N}$, we denote by $\mathcal{B}^r$ the set of test-function $\phi \in \mathscr{D}$ supported in the unit ball $B(0,1)$ and such that $\|\phi\|_{C^r} \leq 1$.

Définition 2.2 (Scaling)**.** For $\phi \in \mathscr{D}, \lambda \in \mathbb{R}, x \in \mathbb{R}^d$, we set

$$\forall y \in \mathbb{R}^d, \quad \phi_x^\lambda(y) \coloneqq \lambda^{-d}\phi\left(\frac{y-x}{\lambda}\right)$$

Remarque 2.3. For $\phi, \psi \in \mathscr{D}$, and $x \in \mathbb{R}^d, \lambda > 0$, we have $(\phi * \psi)_x^\lambda = \phi^\lambda * \psi_x^\lambda$.

Indeed,

$$\begin{aligned}
\forall y \in \mathbb{R}^d, (\phi * \psi)_x^\lambda(y) &= \lambda^{-d} \int \mathrm{d}z \phi(z)\psi(\lambda^{-1}(y-x)-z) \\
&= \int \mathrm{d}z \lambda^{-d}\phi(\lambda^{-1}z)\lambda^{-d}\psi(\lambda^{-1}(y-x-z)) \quad \text{(change of variable } z \leftarrow \lambda z) \\
&= \int \mathrm{d}z \phi^\lambda(z)\psi_x^\lambda(y-z).
\end{aligned}$$

Another operation on distribution that will be very useful is the convolution by a test-function.

Définition 2.4. Let $T \in \mathscr{D}'$ and $\rho \in \mathscr{D}$. Define the function $T * \rho$ by :

$$\forall x \in \mathbb{R}^d, \quad (T * \rho)(x) = T(\rho(x - \cdot)).$$

Proposition 2.5. *With the same notation as in the definition, $T * \rho$ is smooth and for all $k \in \mathbb{N}^d$,*

$$D^k(T * \rho) = (D^k T) * \rho = T * (D^k \rho)$$

Proof. The second equality holds by definiton of the derivative of a distribution. So to prove the proposition, it suffices to show that $T * \rho$ is differentiable and that for $i \in \{1, \dots, d\}, \quad \partial^i(T * \rho) = T * (\partial^i)$.

$$\forall x, y \in \mathbb{R}^d, \quad T * \rho(y) - T * \rho(x) - \sum i = 1d(y_i - x_i)T * (\partial^i \rho)(x) = T(\eta_{x,y}),$$

where $\eta_{x,y} = \rho(y - \cdot) - \rho(x - \cdot) - \sum_{i=1}^d (y_i - x_i)\partial^i \rho(x - \cdot)$. Fix $x \in \mathbb{R}^d$ and $R > 0$. There exists a compact K such that for all $y \in B(x, R)$, the support of $\eta_{x,y}$ is included in K. So by definition of a distribution, there exists $r \in \mathbb{N}$ such that :

$$\forall y \in B(x, R), \quad |T(\eta_{x,y})| \lesssim \|\eta_{x,y}\|.$$

Now by Taylor's theorem, $\|\eta_{x,y}\| = o(x-y)$, so $T*\rho$ is differentiable and $\partial^i(T*\rho) = T*(\partial^i \rho)$ for $i \in \{1, \dots, d\}$. □

Lemme 2.6. *Let $\phi \in D$ be a test-function. $T * \rho$ viewed as a distribution ($C^\infty \subset L^\infty_{loc}$),*

$$(T * \rho)(\phi) = T(\tilde{\rho} * \phi),$$

where $\tilde{\rho} = \rho(-\cdot)$.

Proof. Fix $a < b \in \mathbb{R}$ such that $\mathrm{Supp}(\phi) \subset [a, b]^d$. Let $\delta_n := \frac{b-a}{2^n}$ and $\mathcal{P}_n := \{a + i\delta_n | i = 0, \dots, 2^n - 1\}$.

$$\begin{aligned}
(T * \rho)(\phi) &= \int (T * \rho)(x)\phi(x) \,\mathrm{d}x \\
&= \int T(\rho(x - \cdot))\phi(x) \,\mathrm{d}x \\
&= \int_{[a,b]^d} T(\rho(x - \cdot)\phi(x)) \,\mathrm{d}x \\
&= \lim_{n \to +\infty} T(\underbrace{\sum_{x \in \mathcal{P}_n^d} \delta_n^d \rho(x - \cdot)\phi(x))}_{I_n(\cdot)} \quad \text{(Riemann sum)} \\
&= T(\lim_{n \to +\infty} \sum_{x \in \mathcal{P}_n^d} \delta_n \rho(x - \cdot)\phi(x)) \\
&= T(\int \rho(x - \cdot)\phi(x) \,\mathrm{d}x) \\
&= T(\tilde{\rho} * \phi).
\end{aligned}$$

The exchange $\lim_n T(I_n) = T(\lim_n I_n)$ is justified by the facts that :

- because $[a,b]^d$ and $\mathrm{Supp}(\eta)$ are compact, there exists a compact K such that for all $n \in \mathbb{N}$, $\mathrm{Supp}(\sum_{x\in\mathcal{P}_n^d} \delta_n^d \rho(x-\cdot)\phi(x)) \subset K$,
- by definition of a distribution, there exists $r \in \mathbb{N}$ such that T is continuous on $C^\infty(K)$ equiped with $\|\cdot\|_{C^r}$,
- finally, $\sum_{x\in\mathcal{P}_n^d} \delta_n^d \rho(x-\cdot)\phi(x) \xrightarrow[n\to+\infty]{\|\cdot\|_{C^r}} \int \rho(x-\cdot)\phi(x)\,\mathrm{d}x$ by the Riemann sum property on the successive derivatives.

$\square$

Théorème 2.7 (Mollification)**.** *Let $T \in \mathscr{D}'$ be a distribution, and $\rho \in \mathscr{D}$ be any test-function such that $\int \rho = 1$. Then, for any test-function $\phi \in \mathscr{D}$, and all $r \in \mathbb{N}$,*

$$\left\|\phi - \rho^\lambda * \phi\right\|_{C^r} \xrightarrow[\lambda\to 0]{} 0,$$

and consequently,

$$T * \rho^\lambda(\phi) \xrightarrow[\lambda\to 0]{} T(\phi).$$

First, let $r \in \mathbb{N}$ and $k \in \mathbb{N}^d$ such that $|k| \leq r$. Differentiating under the integral sign, and knowing $\int \rho = 1$, we have :

$$\partial^k(\phi - \rho^\lambda * \phi)(x) = \int \rho^\lambda(z)(\partial^k\phi(x) - \partial^k\phi(z-x))\,\mathrm{d}z.$$

So that by Taylor's theorem,

$$\begin{aligned}\left\|\phi - \rho^\lambda * \phi\right\|_{C^r} &\leq \|\phi\|_{C^{r+1}} \int \rho^\lambda(z)\,|z|\,\mathrm{d}z \\ &\leq \lambda\,\|\phi\|_{C^{r+1}} \int \rho(u)\,|u|\,\mathrm{d}u.\end{aligned}$$

Now for the second statement, we proved that $T * \rho^\lambda(\phi) = T(\tilde{\rho}^\lambda * \phi)$ and $\tilde{\rho}^\lambda * \phi \xrightarrow[\lambda\to 0]{\|\cdot\|_{C^r}} \phi$. Additionnaly, we can find a compact K such that for all $\lambda \in (0,1]$, $\quad \mathrm{Supp}(\tilde{\rho}^\lambda * \phi) \subset K$, So as before, the continuity property of T applies.

Définition 2.8 (α-Hölder functions)**.** Let $\alpha \in (0,1)$. We say that a function f is α-Hölder if

$$\forall x, y \in \mathbb{R}^d, |f(x) - f(y)| \lesssim |x-y|^\alpha.$$

We say that f is locally α-Hölder if the constant in the estimate depends on the compact in which we take x and y. For $\alpha \in \mathbb{R}_+ \setminus \mathbb{N}$, we say that f is α-Hölder if f is $\lfloor\alpha\rfloor$-times differentiable, and

$$\|f\|_\alpha \coloneqq \|f\|_\infty + [f]_\gamma < \infty$$

where

$$[f]_\gamma \coloneqq \sup_{x\in\mathbb{R}^d} \sup_{h\in\mathbb{R}^d} \frac{\left|f(x+h) - \sum_{|k|<\gamma} \frac{h^k}{k!}\partial^k f(x)\right|}{|h|^\gamma}.$$

We can also define Hölder spaces for negative α.

First, given a positive integer r, note B_r the set of smooth functions ϕ supported in the unit ball, with :

$$\|\phi\|_{C^r} := \sup_{|k|\leq r} \left\|D^k\phi\right\|_\infty \leq 1.$$

Définition 2.9 (Negative Hölder spaces)**.** Let $\alpha < 0$. T is α-Hölder if

$$\forall \phi \in B_r, \quad \left|T(\phi_x^\lambda)\right| \lesssim |\lambda|^\alpha \quad \text{uniformly in } x \in \mathbb{R}^d, \lambda \in (0,1)$$

where $r > \lceil -\alpha \rceil$. We can also talk about local α-Hölder distribution, when the estimate is uniform in $x \in K$ for all K compact.

It will be useful to further generalise this notion by relaxing the uniform control in x and λ, asking for

$$\left\| \left\| \sup_{\phi \in B_r} \frac{T(\phi_x^\lambda)}{|\lambda|^\alpha} \right\|_{L^p(x\in\mathbb{R}^d)} \right\|_{L^q(\lambda\in(0,1))} < \infty$$

This defines the Besov space $\mathcal{B}^\alpha_{p,q}$.

Remarque 2.10.

- We have $\mathcal{B}^\alpha_{p,q_1} \hookrightarrow \mathcal{B}^\alpha_{p,q_2}$ for $q_1 \leq q_2$.
- We have $\mathcal{B}^\alpha_{p,q} \hookrightarrow \mathcal{B}^\beta_{p,q}$ for $\beta \leq \alpha$ on compact domains.

A very useful lemma allows us to check the finiteness of this scaling norm only for one compactly supported test function ϕ with $\int \phi \neq 0$. and for λ on the dyadic scale.

Lemme 2.11. *Let $\alpha < 0$. Then $T \in \mathcal{B}^\alpha_{p,q}$ if and only if there exists a compactly supported test-function ϕ with $\int \phi \neq 0$ and*

$$\left\| \left\| \frac{T(\phi_x^{2^{-n}})}{2^{-n\alpha}} \right\|_{L^p(x\in\mathbb{R}^d)} \right\|_{l^q(n)} < \infty.$$

We present the proof because it displays ideas that will reappear later, in reconstruction arguments for instance.

Proof. The direct implication is clear. Fix $T \in D'$ and ϕ such that the statement holds, and let's prove that $T \in \mathcal{B}^\alpha_{p,q}$. Fix $r \in \mathbb{N}$ such that $r > -\alpha$, $\lambda_0, \dots, \lambda_{r-1} \in \mathbb{R}_+^*$ distincts, $c_0, \dots, c_{r-1} \in \mathbb{R}$, and

$$\hat{\phi} := \frac{1}{\int \phi} \sum_{i=0}^{r-1} c_i \phi^{\lambda_i}$$

We can tune the constants c_i so that

- $\int \hat{\phi} = 1$,
- $\forall k, 1 \leq |k| \leq r-1, \quad \int x^k \hat{\phi}(x)\,\mathrm{d}x = 0$,

Indeed,

$$\begin{aligned}
\forall k \in \mathbb{N}^d, |k| \leq r-1, \int x^k \hat{\phi}(x)\,\mathrm{d}x = \delta_{0,k} &\iff \frac{1}{\int \phi} \sum_{i=0}^{r-1} c_i \int x^k \phi^{\lambda_i}(x)\,\mathrm{d}x = \delta_{0,k} \\
&\iff \frac{1}{\int \phi} \sum_{i=0}^{r-1} c_i \lambda_i^{|k|} \int \phi = \delta_{0,k} \quad \text{(change of variable)} \\
&\iff \sum_{i=0}^{r-1} c_i \lambda_i^{|k|} = \delta_{0,k}
\end{aligned}$$

Because the λ_i are distincts, this is an inversible Vandermonde system. Additionnaly, by choosing small λ_is, we see that we can shrink the support of of $p\hat{h}i$ so that it is included in $B(0, \frac{1}{2})$.

Let $\lambda_i = 2^{m_i}$ for $m_i \in \mathbb{N}$ large enough.

$$\begin{aligned}
\left\| \left\| \frac{T(\hat{\phi}_x^{2^{-n}})}{2^{-n\alpha}} \right\|_{L^p(x \in \mathbb{R}^d)} \right\|_{l^q(n)} &\lesssim \sum_{i=0}^{r-1} \left\| \left\| \frac{T(\phi_x^{2^{-n-m_i}})}{2^{-n\alpha}} \right\|_{L^p(x \in \mathbb{R}^d)} \right\|_{l^q(n)} \qquad \text{(triangle inequality)} \\
&\lesssim \sum_{i=0}^{r-1} \left\| \left\| \frac{T(\phi_x^{2^{-n-m_i}})}{2^{-(n+m_i)\alpha}} \right\|_{L^p(x \in \mathbb{R}^d)} \right\|_{l^q(n)} \\
&< \infty
\end{aligned}$$

Let $\psi \in D'$ be any test function,and ρ a mollifier with $\int \rho = 1$. We know that $\int T(\rho_z^{2^{-m}})\psi(z)\,\mathrm{d}z = T(\rho^{2^{-m}} * \psi) \xrightarrow[m \to \infty]{} T(\psi)$. So we can write the telescopic sum :

$$T(\psi_x^{2^{-n}}) = \int T(\rho_z^{2^{-n}})\psi_x^{2^{-n}}(z)\,\mathrm{d}z + \sum_{m \geq n} \int T((\rho^{2^{-m-1}} - \rho^{2^{-m}})_z)\psi_x^{2^{-n}}(z)\,\mathrm{d}z$$

Now we use $\rho := \hat{\phi}^2 * \hat{\phi}$, motivated by the fact that we can write the increment in convolution form : $\rho^{2^{-m-1}} - \rho^{2^{-m}} = (\rho^{\frac{1}{2}} - \rho)^{2^{-m}} = \hat{\phi}^{2^{-m}} * \check{\phi}^{2^{-m}}$ where $\check{\phi} = \hat{\phi}^{\frac{1}{2}} - \hat{\phi}^2$. Plugging in the equation :

$$\begin{aligned}
T(\psi_x^{2^{-n}}) &= \int T(\hat{\phi}^{2^{-n}} * \hat{\phi}_z^{2^{-n+1}})\psi_x^{2^{-n}}(z)\,\mathrm{d}z + \sum_{m \geq n} \int T(\hat{\phi}^{2^{-m}} * \check{\phi}_z^{2^{-m}})\psi_x^{2^{-n}}(z)\,\mathrm{d}z \\
&= \int T(\hat{\phi}_z^{2^{-n}})(\hat{\phi}^{2^{-n+1}} * \psi^{2^{-n}})(z-x)\,\mathrm{d}z + \sum_{m \geq n} \int T(\hat{\phi}_z^{2^{-m}})(\check{\phi}^{2^{-m}} * \psi^{2^{-n}})(z-x)\,\mathrm{d}z \\
&= \int_{B(0,2^{-n+1})} T(\hat{\phi}_{y+x}^{2^{-n}})(\hat{\phi}^{2^{-n+1}} * \psi^{2^{-n}})(y)\,\mathrm{d}y + \sum_{m \geq n} \int_{B(0,2^{-n+1})} T(\hat{\phi}_{y+x}^{2^{-m}})(\check{\phi}^{2^{-m}} * \psi^{2^{-n}})(y)\,\mathrm{d}y
\end{aligned}$$

The last line is justified by the fact that $y = z - x$, and that $y \mapsto T(\hat{\phi}_{y+x}^{2^{-n}})$ is supported in $B(0, 2^{-n+1})$. To continue, we need the following lemma :

Lemme 2.12. *Assume that $\nu \in D$ is a test function that cancels polynomials up to degree $r - 1 \in \mathbb{N}$: $\forall 0 \leq k \leq r-1, \int x^k \nu(x)\,\mathrm{d}x = 0$. Then for any test function $\eta \in D$ and $\lambda > 0$,*

$$\left\| \nu^\lambda * \eta \right\|_\infty \lesssim \|\eta\|_{C^r} \|\nu\|_{L^1} \lambda^r$$

where the constant depends on the support of ν and r.

Proof. Fix $\eta \in D$ and $\lambda > 0$. Denote for all $x, y \in \mathbb{R}^d$, $\quad T_x^{r-1}\eta(y) = \sum_{|k| \leq r-1} \frac{D^k\eta(x)}{k!}(y-x)^k$ the Taylor polynomial at x up to order $r-1$ of η.

By Taylor's theorem, $|(\eta - T_x^r\eta)(y)| \leq \frac{\|\eta\|_{C^r}}{r!} |y - x|^r$.

$$\forall x \in \mathbb{R}^d, \quad \left|\nu^\lambda * \eta(x)\right| \leq \left| \int \nu^\lambda(y)(\eta - T_x^r \eta)(x-y)\,\mathrm{d}y + \underbrace{\int \nu^\lambda(y)(T_x^r\eta)(x-y)\,\mathrm{d}y}_{=0 \text{ by hypothesis}} \right|$$
$$\leq \int \left|\nu^\lambda(y)\right| \left|(\eta - T_x^r\eta)(x-y)\right|\,\mathrm{d}y$$
$$\leq \int \left|\nu^\lambda(y)\right| \frac{\|\eta\|_{C^r}}{r!} |y|^r\,\mathrm{d}y$$
$$\leq \frac{\|\eta\|_{C^r}}{r!}\lambda^r \int_K |\nu(z)|\,|z|^r\,\mathrm{d}z$$
$$\leq \frac{1}{r!}(\sup_{z\in K}|z|^r)\,\|\eta\|_{C^r}\,\|\nu\|_{L^1}\,\lambda^r \quad (\text{uniform bound in } x)$$

where K denotes the (compact) support of ν. □

Thanks to this, we have :

$$\begin{cases} \left\|\hat{\phi}^{2^{-n+1}} * \psi^{2^{-n}}\right\|_\infty \lesssim 2^{nd}\,\|\psi\|_\infty \left\|\hat{\phi}\right\|_{L^1} \leq 2^{nd}\,\|\psi\|_{C^r} \left\|\hat{\phi}\right\|_{L^1} \\ \left\|\check{\phi}^{2^{-m}} * \psi^{2^{-n}}\right\|_\infty \lesssim 2^{nd}2^{(n-m)r}\,\|\psi\|_{C^r} \left\|\check{\phi}\right\|_{L^1} \end{cases}$$

As a consequence, denoting $B_n = B(0, 2^{-n})$,

$$\sup_{\psi\in B^r} \left|T(\psi_x^{2^{-n}})\right| \lesssim 2^{nd} \left\|\hat{\phi}\right\|_{L^1} \int_{B_{n-1}} T(\hat{\phi}_{y+x}^{2^{-n}})\,\mathrm{d}y + \sum_{m\geq n} 2^{nd+(n-m)r} \left\|\check{\phi}\right\|_{L^1} \int_{B_{n-1}} T(\check{\phi}_{y+x}^{2^{-m}})\,\mathrm{d}y.$$

Taking L^p-norm in x, we get by triangle inequality :

$$\left\|\sup_{\psi\in B^r}\left|T(\psi_x^{2^{-n}})\right|\right\|_{L^p(x)} \lesssim 2^{nd}\left\|\hat{\phi}\right\|_{L^1}\int_{B_{n-1}}\left\|T(\hat{\phi}_{y+x}^{2^{-n}})\right\|_{L^p(x)}\mathrm{d}y + \sum_{m\geq n} 2^{nd+(n-m)r}\left\|\check{\phi}\right\|_{L^1}\int_{B_{n-1}}\left\|T(\hat{\phi}_{y+x}^{2^{-m}})\right\|_{L^p(x)}\mathrm{d}y$$

Now in these integrals, the integrand is actually constant, we get :

$$\left\|\sup_{\psi\in B^r}\left|T(\psi_x^{2^{-n}})\right|\right\|_{L^p(x)} \lesssim \left\|T(\hat{\phi}_x^{2^{-n}})\right\|_{L^p(x)} + \sum_{m\geq n} 2^{(n-m)r}\left\|T(\hat{\phi}_x^{2^{-m}})\right\|_{L^p(x)}.$$

Then, dividing by $2^{-n\alpha}$ and taking the l^q-norm in n,

$$\|T\|_{\mathcal{B}^\alpha_{p,q}} \lesssim \left\|\left\|\frac{T(\hat{\phi}_x^{2^{-n}})}{2^{-n\alpha}}\right\|_{L^p(x)}\right\|_{l^q(n)} + \left\|\sum_{m\geq n} 2^{(n-m)(r+\alpha)}\left\|\frac{T(\hat{\phi}_x^{2^{-m}})}{2^{-m\alpha}}\right\|_{L^p(x)}\right\|_{l^q(n)}.$$

And finally, because we chose $r > -\alpha$, applying Jensen's inequality, then using Fubini for the sums in n and m, we proved that $\|T\|_{\mathcal{B}^\alpha_{p,q}} < \infty$.

□

Juste like with Sobolev spaces with Sobolev embeddings, we can trade regularity for integrability via the following theorem.

Théorème 2.13. *Let $q > 0$, $0 < p_0 \leq p_1 \leq \infty$, and $-\infty < s_1 \leq s_0 < \infty$. Then,*

$$s_0 - \frac{n}{p_0} = s_1 - \frac{n}{p_1} \implies \mathcal{B}^{s_0}_{p_0,q} \hookrightarrow \mathcal{B}^{s_1}_{p_1,q}.$$

See [5] for a general proof. We are only going to use this in a specific case where the proof will be easier.

We said that we could not multiply two distributions in general, but Hölder regularity gives a simple sufficient condition.

2.2 Random distributions and noise

Définition 2.14 (Random distributions)**.** A random distribution T is a map $(\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow \mathscr{D}'$ such that for all test-function $\phi \in \mathscr{D}$, $T(\phi) \coloneqq T \circ \mathrm{ev}_\phi : (\Omega, \mathcal{F}, \mathbb{P}) \longrightarrow \mathbb{R}$ is measurable (i.e. a real random variable).

The forcing term ξ in our equation is a random distribution. A common choice is white noise, let us describe it's law.

2.3 Germs and reconstruction

We return to our problem of defining the product between a distribution and a function. Assume that our function f has (positive) regularity α, and our distribution g negative regularity β. If it is not clear how we could define the product fg globally, we know that around a point x, f is well-approximated by its Taylor polynomial at x : $T_x^\alpha f = \sum_{|k|<\alpha} \frac{f^{(k)(x)}}{k!}(\cdot - x)^k$. Of course as a polynomial, $T_x^\alpha f$ is smooth, so the product $(T_x^\alpha f)g$ is well-defined, and is expected to approximate the formal fg around x. The way of defining fg will be to reconstruct a global distribution from the local approximations $T_x^\alpha f$, for $x \in \mathbb{R}^d$.

Définition 2.15 (Germs of distribution)**.** A germ of distribution is a family of distribution $(F_x)_{x\in\mathbb{R}^d}$ indexed by our space variable. Additionnaly, we require for technical reason that for all test-function $\phi \in \mathscr{D}$, the map $x \mapsto F_x(\phi)$ should be measurable.

Exemple 2.16. $((T_x^\alpha f)g)_{x\in\mathbb{R}^d}$ is a germ of distributions.

Théorème 2.17 (Reconstruction)**.** *Let $(F_x)_{x\in\mathbb{R}^d}$ be a germ. Let $\gamma > 0$, and assume that for some $\phi \in \mathscr{D}$ with $\int \phi \neq 0$ and some $\alpha < \gamma$, we have the coherence condition :*

$$\|F\|_{\alpha,\gamma;\phi} \coloneqq \sup_{x,y\in\mathbb{R}^d} \sup_{\lambda>0} \frac{\left|(F_x - F_y)(\phi_x^\lambda)\right|}{\lambda^\alpha(\lambda + |x-y|)^{\gamma-\alpha}} < \infty. \tag{2.1}$$

Then, there exists a unique distribution $\mathcal{R}(F) \in \mathscr{D}'$ called the reconstruction of F, that is locally well approximated by the germ :

$$\sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\psi\in\mathcal{B}^r} \frac{\left|(\mathcal{R}(F) - F_x)(\psi_x^\lambda)\right|}{\lambda^\gamma} \lesssim \|F\|_{\alpha,\gamma;\eta},$$

with $r \in \mathbb{N}, r > -\alpha$. The implicit constant depending only on α, γ, r, ϕ and d.

Remarque 2.18.
- The coherence condition (2.1) can be interpreted as follows : the regularity of the increments $F_x - F_y$ is α, and when setting $|x-y| \leq \lambda \to 0$, the increment tested on η_x^λ behaves like λ^γ. This quantity goes to 0 because $\gamma > 0$, this is a continuity condition on $x \mapsto F_x$.
- This is one version of a reconstruction theorem, in negative Hölder space, but we will present another example in Besov spaces.

- If we have also an homogeneity bound on the germ, meaning there exists $\beta \in \mathbb{R},\ \alpha < \beta \leq \gamma$ such that $\|F\|_{\mathrm{coh},\beta} \coloneqq \sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\psi\in\mathcal{B}^r} \frac{\left|F_x(\psi_x^\lambda)\right|}{\lambda^\beta} < \infty$, then by triangle inequality, we see that :

$$\sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\psi\in\mathcal{B}^r} \frac{\left|\mathcal{R}(F)(\psi_x^\lambda)\right|}{\lambda^\beta} \leq \|F\|_{\mathrm{hom},\beta} + \|F\|_{\alpha,\gamma;\phi} < \infty.$$

Thus the reconstruction is $\beta \wedge \gamma$-Hölder.

We return to the problem of mutliplying a distribution with a non-smooth function.

Corollaire 2.19 (Young's multiplication)**.** *Let $\alpha \in \mathbb{R}_+^* \setminus \mathbb{N}$, $\beta < 0$ be such that $\alpha + \beta > 0$. Then, there exists a unique bilinear map $C^\alpha \times C^\beta \longrightarrow C^\alpha$, that extends the multiplication operator $C^\infty \times \mathscr{D}' \longrightarrow \mathscr{D}'$.*

Proof. Let $f \in C^\alpha$ and $g \in C^\beta$. Recall that we introduced the germ $(F_x \coloneqq (T_x^\alpha f)g)_{x\in\mathbb{R}^d}$ and that we aimed to solve the problem of multiplication by applying reconstruction to it. We will check that this germ is $(\beta, \alpha+\beta)$-coherent :

$$\begin{aligned}
\forall x, y \in R^d, \quad F_x - F_y &= \left(\sum_{|k|<\alpha} \frac{D^k f(x)}{k!} (\cdot - x)^k - \sum_{|k|<\alpha} \frac{D^k f(y)}{k!} (\cdot - y)^k \right) g \\
&= \left(\sum_{|k|<\alpha} \frac{D^k f(x)}{k!} (\cdot - x)^k - \sum_{|k|<\alpha} \frac{D^k f(y)}{k!} \sum_{|l|\leq|k|} \binom{k}{l} (\cdot - x)^l (x-y)^{k-l} \right) g \\
&= \left(\sum_{|k|<\alpha} \left(\frac{D^k f(x)}{k!} - \sum_{|k|\leq|j|<\alpha} \frac{D^j f(y)}{j!} \binom{j}{k} (x-y)^{j-k} \right) (\cdot - x)^k \right) g \\
&= \left(\sum_{|k|<\alpha} \frac{1}{k!} \left(D^k f(x) - \sum_{|j|<\alpha-|k|} \frac{D^{j+k} f(y)}{j!} (x-y)^j \right) (\cdot - x)^k \right) g
\end{aligned}$$

Now f is in C^α, so for k such that $|k| < \alpha$, $D^k f$ is in $C^{\alpha-|k|}$, and

$$\forall x, y \in \mathbb{R}^d, \qquad \left| D^k f(x) - \sum_{|j|<\alpha-|k|} \frac{D^{j+k} f(y)}{j!} (x-y)^j \right| \leq [f]_\alpha \left|x-y\right|^{\alpha-|k|}.$$

As a result, for any test-function $\eta \in \mathscr{D}$, $\lambda > 0$ and $x, y \in \mathbb{R}^d$,

$$\begin{aligned}
\left|(F_y - F_x)(\eta_x^\lambda)\right| &\leq \sum_{|k|<\alpha} \frac{1}{k!} [f]_\alpha \left|x-y\right|^{\alpha-|k|} \left|g((\cdot - x)^k \eta_x^\lambda)\right| \\
&\leq \sum_{|k|<\alpha} \frac{1}{k!} [f]_\alpha \left|x-y\right|^{\alpha-|k|} \lambda^{|k|} \left|g(((\cdot)^k \eta)_x^\lambda)\right| \\
&\leq [f]_\alpha \|g\|_\beta \sum_{|k|<\alpha} \frac{1}{k!} \left|x-y\right|^{\alpha-|k|} \lambda^{\beta+|k|} \quad \text{(by definition of } \beta\text{-Hölder norm)} \\
&\lesssim [f]_\alpha \|g\|_\beta \lambda^\beta (\lambda + |x-y|)^{\alpha+\beta}.
\end{aligned}$$

$\alpha + \beta > 0$ so we can apply the reconstruction theorem : there exists a unique $fg \coloneqq \mathcal{R}(F) \in \mathscr{D}'$, such that

$$\sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\phi\in\mathcal{B}^r} \frac{\left|(fg - (T_x^\alpha f)g)(\phi_x^\lambda)\right|}{\lambda^{\alpha+\beta}} \lesssim \|F\|_{\alpha,\gamma;\eta}$$

Also,

$$\begin{aligned}
\left|(F_x)(\eta_x^\lambda)\right| &\leq \sum_{|k|<\alpha} \left|\frac{D^k f(x)}{k!}\right| \left|g((\cdot - x)^k \eta_x^\lambda)\right| \\
&\leq \sum_{|k|<\alpha} \left|\frac{D^k f(x)}{k!}\right| \lambda^{|k|} \left|g(((\cdot)^k \eta)_x^\lambda)\right| \\
&\leq \|g\|_\beta \sum_{|k|<\alpha} \left|\frac{D^k f(x)}{k!}\right| \lambda^{|k|+\beta} \\
&\lesssim \|g\|_\beta \|f\|_{C^\alpha} \lambda^{\alpha+\beta} \quad \text{uniformly in } \eta,\ x \text{ and } \lambda.
\end{aligned}$$

So by the third item in the remark above, fg is $\alpha+\beta$-Hölder. Additionnaly, the map, $(f,g) \mapsto (T_x^\alpha f)g$ is bilinear, and $\mathcal{R} : F_x \mapsto \mathcal{R}(F)$ is linear (by easily seen by uniqueness). So $(f,g) \mapsto fg$ is bilinear. It remains to check that it extends the multiplication map. For this, it is enough to check that the product verifies the reconstruction bound when $f \in C^\infty$.

Suppose $f \in C^\infty$.

$$\begin{aligned}
\forall x \in \mathbb{R}^d, \lambda > 0, \phi \in \mathcal{B}^r, \left|(fg - (T_x^\alpha f)g)(\phi_x^\lambda)\right| &= \left|g((f - (T_x^\alpha f))(\phi_x^\lambda))\right| \\
&= \left|g(((\sum_{|k|=\lfloor\alpha\rfloor+1} R_k(\cdot)(\cdot - x)^k))(\phi_x^\lambda))\right| \\
&= \lambda^{\lfloor\alpha\rfloor+1} \left|g((((\sum_{|k|=\lfloor\alpha\rfloor+1} R_k(\cdot + x)^{\lambda^{-1}}(\cdot)^k))(\phi))_x^\lambda)\right| \\
&\lesssim \lambda^{\alpha+\beta} \quad (g \in C^\beta)
\end{aligned}$$

The second line is justified by Taylor's theorem because f is smooth, $R_k(\cdot)$ is smooth and vanishes at x. □

Proof. Fix $(F_x)_{x\in\mathbb{R}^d}$ and ϕ as the statement of the theorem.

Step 1 : Uniqueness

Let R_1 and R_2 be two distributions such that the reconstruction bound holds :

$$\sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\psi\in\mathcal{B}^r} \frac{\left|(R_i - F_x)(\psi_x^\lambda)\right|}{\lambda^\gamma} \lesssim \|F\|_{\alpha,\gamma;\phi}, \quad i = 1,2.$$

Setting $R = R_1 - R_2$, by triangle inequality, we have :

$$\sup_{x\in\mathbb{R}^d} \sup_{\lambda>0} \sup_{\psi\in\mathcal{B}^r} \frac{\left|R(\psi_x^\lambda)\right|}{\lambda^\gamma} < \infty.$$

In particular, because $\gamma > 0$, for $\psi \in \mathcal{B}^r, \int \psi = 1$ a mollifier, $\sup_{x\in\mathbb{R}^d} \left|R(\phi_x^\lambda)\right| \xrightarrow[\lambda\to 0]{} 0$.

Let $\eta \in \mathscr{D}$ be any test-function.

$$\begin{aligned}
R(\eta) &= \lim_{\lambda\to 0} \int R(\psi_x^\lambda)\eta(x)\,\mathrm{d}x \quad \text{(mollification)} \\
&= 0 \quad \text{(by uniform convergence)}
\end{aligned}$$

So we proved that $R = 0$.

Step 2 : Existence

We want to construct a distribution well-approximated by F_x close to x. Let $\rho \in \mathscr{D}$ be a test-function such that $\int \rho = 1$, to be specified later. By analogy with the mollification property, we want to define for all $n \in \mathbb{N}$ and $\eta \in \mathscr{D}$, $R_n(\eta) \coloneqq \int F_x * \rho^{2^{-n}}(x)\eta(x)\,\mathrm{d}x$ and hope to show that R_n has a limit when n goes to $+\infty$.

The function $R_n : x \mapsto F_x * \rho^{2^{-n}}(x)$ is well-defined and measurable. Indeed, for all $y \in \mathbb{R}^d$, $\quad x \mapsto F_x * \rho^{2^{-n}}(y) = F_x(\rho^{2^{-n}}(y - \cdot))$ is measurable by definition of a germ, and for all $x \in \mathbb{R}^d$, $\quad y \mapsto F_x * \rho^{2^{-n}}(y)$ is continuous by the mollification property.

We want to estimate the difference $\mathcal{R}(F) - F_x$ so it is natural to introduce the mollified version :

$$\forall x, z \in \mathbb{R}^d,\ n \in \mathbb{N}, \quad f_{x,n}(z) = R_n(z) - F_x * \rho^{2^{-n}}(z) = (F_z - F_x) * \rho^{2^{-n}}(z).$$

Fix $x \in \mathbb{R}^d$. $f_{x,n}$ is a measurable function. We want to show that it converges to a distribution when n goes to $+\infty$.

$$\forall z \in \mathbb{R}^d, \quad (f_{x,n+1} - f_{x,n})(z) = (F_z - F_x) * (\rho^{2^{-n-1}} - \rho^{2^{-n}})(z)$$

Just as in the proof of the technical lemma 2.11, we define $\hat{\phi} \coloneqq \frac{1}{\int \phi} \sum_{i=0}^{r-1} c_i \phi^{\lambda_i}$ for suitable constants (c_i), (λ_i), and $\rho \coloneqq \hat{\phi}^2 * \hat{\phi}$, so that we can write the increment in convolution form : $\rho^{2^{-m-1}} - \rho^{2^{-m}} = (\rho^{\frac{1}{2}} - \rho)^{2^{-m}} = \hat{\phi}^{2^{-m}} * \check{\phi}^{2^{-m}}$ where $\check{\phi} \coloneqq \hat{\phi}^{\frac{1}{2}} - \hat{\phi}^2$. Recall that $\int \hat{\phi} = 1$ so that $\int \rho = 1$: we can use it as our mollifier. Additionnaly, $\forall k, 1 \leq |k| \leq r-1, \quad \int x^k \hat{\phi}(x)\,\mathrm{d}x = 0$ where we fixed $r \in N$, $r > -\alpha$, and the same goes trivially for $\check{\phi}$. Finally, recall that the coherence bound is verified with $\hat{\phi}$.

$$\begin{aligned}
\forall z \in \mathbb{R}^d, (f_{x,n+1} - f_{x,n})(z) &= (F_z - F_x) * (\hat{\phi}^{2^{-n}} * \check{\phi}^{2^{-n}})(z) \\
&= (F_z - F_x)((\hat{\phi}^{2^{-n}} * \check{\phi}^{2^{-n}})(z - \cdot)) \\
&= (F_z - F_x) * \tilde{\hat{\phi}}^{2^{-n}}((\check{\phi}^{2^{-n}})(z - \cdot)) \\
&= \int (F_z - F_x)(\hat{\phi}_y^{2^{-n}})(\check{\phi}^{2^{-n}})(z - y)\,\mathrm{d}y \\
&= \int (F_y - F_x)(\hat{\phi}_y^{2^{-n}})(\check{\phi}^{2^{-n}})(z - y)\,\mathrm{d}y \quad (=: g'_{x,n}(z)) \\
&\quad + \int (F_z - F_y)(\hat{\phi}_y^{2^{-n}})(\check{\phi}^{2^{-n}})(z - y)\,\mathrm{d}y \quad (=: g''_{x,n}(z))
\end{aligned}$$

We can estimate the first integral with the coherence bound. Let $\psi \in \mathscr{D}$ be any test-function. Fix a compact K such that $\mathrm{Supp}(\psi) \subset K$. Then $\check{\phi}^{2^{-n}} * \tilde{\psi}(\mathrm{y}) \subset \bar{K}_1$ where

$\bar{K}_1 \coloneqq \{x \in \mathbb{R}^d | d(x, K) \le 1\}$ is compact. We can compute :

$$\begin{aligned}
\left|g'_{x,n}(\psi)\right| &= \left|\int \mathrm{d}z\psi(z) \int (F_y - F_x)(\hat{\phi}_y^{2^{-n}})(\check{\phi}^{2^{-n}})(z-y)\,\mathrm{d}y\right| \\
&= \left|\int \mathrm{d}y(F_y - F_x)(\hat{\phi}_y^{2^{-n}})(\check{\phi}^{2^{-n}}) * \tilde{\psi}(y)\right| \quad \text{(Fubini, justified a posteriori.)} \\
&\lesssim \|F\|_{\alpha,\gamma;\phi} \int_{\bar{K}_1} \mathrm{d}y 2^{-n\alpha}(2^{-n} + |x-y|)^{\gamma-\alpha} \left|\check{\phi}^{2^{-n}} * \tilde{\psi}(y)\right| \\
&\lesssim 2^{-n\alpha} \|F\|_{\alpha,\gamma;\phi} \left\|\check{\phi}^{2^{-n}} * \tilde{\psi}\right\|_\infty \\
&\lesssim 2^{-n\alpha}2^{-nr} \|F\|_{\alpha,\gamma;\phi} \|\psi\|_{C^r} < \infty \quad \text{by lemma 2.12}
\end{aligned}$$

The finiteness of the left-hand side justifies a posteriori the computation of $g'_{x,n}(\psi)$ which didn't make sense a priori because we didn't prove it was locally integrable.

The second term is easier because $(\check{\phi}^{2^{-n}})(z-y)$ forces $|z-y| \le 2^{-}n$:

$$\begin{aligned}
\forall z \in \mathbb{R}^d, \quad \left|g'_{x,n}(z)\right| &\le \left\|\check{\phi}\right\|_{L^1} \sup_{|z-y|\le 2^{-n}} \left|(F_z - F_y)(\hat{\phi}_y^{2^{-n}})\right| \\
&\lesssim 2^{-n\alpha}(2 \times 2^{-n})^{\gamma-\alpha} \|F\|_{\alpha,\gamma;\phi} \\
&\lesssim 2^{-n\gamma} \|F\|_{\alpha,\gamma;\phi} \quad \text{(uniformly in z).}
\end{aligned}$$

In turn, we showed that $f_{x,n+1} - f_{x,n}$ is locally integrable and that for any test-function $\psi \in \mathscr{D}$ and K compact such that $\mathrm{Supp}(\psi) \subset K$:

$$|(f_{x,n+1} - f_{x,n})(\psi)| \lesssim \|F\|_{\alpha,\gamma;\phi} \|\psi\|_{C^r} (2^{-n(\alpha+r)} + 2^{-n\gamma})$$

As a result, because $\gamma, r + \alpha > 0$, $f_x = f_{x,l} + \sum_{n=l}^{+\infty}(f_{x,n+1} - f_{x,n})$ is a well-defined distribution of order r (the formula is true for any $l \in \mathbb{N}$). And obviously $(f_{x,k})_k$ converges to f_x. Recall that $R_n(z) = f_{x,n}(z) + F_x * \rho^{2^{-n}}(z)$. By the mollication lemma, $z \mapsto F_x * \rho^{2^{-n}}(z)$ is a smooth function, converging as a distribution to F_x. So R_n is a locally integrable function converging as a distribution to a limit noted $\mathcal{R}(F)$. We have the formula :

$$\forall l \in \mathbb{N}, \quad \mathcal{R}(F) = F_x + f_{x,l} + \sum_{n=l}^{+\infty}(f_{x,n+1} - f_{x,n}).$$

We still have to prove the reconstruction bound. Let $\psi \in B^r$, $\lambda > 0$ and $l \in \mathbb{N}$,

$$\begin{aligned}
f_{x,l}(\psi_x^\lambda) &= \int \mathrm{d}z f_{x,l}(z)\psi^\lambda(z-x) \\
&= \int\int \mathrm{d}z\,\mathrm{d}y(F_z - F_x) * (\hat{\phi}^{2^{-l}})(y)\hat{\phi}^{2^{-l+1}}(z-y)\psi^\lambda(z-x)
\end{aligned}$$

So by the coherence bound, for all $l \in \mathbb{N}$ such that $l \ge \log_2(\lambda)$,

$$\begin{aligned}
\left|f_{x,l}(\psi_x^\lambda)\right| &\lesssim \|F\|_{\alpha,\gamma;\phi} \int\int \mathrm{d}z\,\mathrm{d}y 2^{-l\alpha}(2^{-l} + |x-z|)^{\gamma-\alpha} \left|\hat{\phi}^{2^{-l+1}}(z-y)\right| \left|\psi^\lambda(z-x)\right| \\
&\lesssim \|F\|_{\alpha,\gamma;\phi} \lambda^\alpha \int\int \mathrm{d}z\,\mathrm{d}y(2^{-l} + |x-z|)^{\gamma-\alpha} \left|\hat{\phi}^{2^{-l+1}}(z-y)\right| \left|\psi^\lambda(z-x)\right|
\end{aligned}$$

$\hat{\phi}^{2^{-l+1}}$ is supported in $B(0, 2^{-l})$, so that on the domain of the integral, $|x - z| \leq \lambda$. As a result :

$$\begin{aligned}
\left| f_{x,l}(\psi_x^\lambda) \right| &\lesssim \|F\|_{\alpha,\gamma;\phi} \lambda^\gamma \int\int \mathrm{d}z \, \mathrm{d}y \left| \hat{\phi}^{2^{-l+1}}(z - y) \right| \left| \psi^\lambda(z - x) \right| \\
&\lesssim \|F\|_{\alpha,\gamma;\phi} \lambda^\gamma \left\| \hat{\phi}^{2^{-l+1}} \right\|_{L^1} \left\| \psi^\lambda \right\|_{L^1} \\
&\lesssim \|F\|_{\alpha,\gamma;\phi} \lambda^\gamma \left\| \hat{\phi} \right\|_{L^1} \|\psi\|_{L^1} \\
&\lesssim \|F\|_{\alpha,\gamma;\phi} \lambda^\gamma
\end{aligned}$$

Taking $l \to +\infty$, we recover the reconstruction bound.

$\square$

Chapter 3

Regularity structures

Définition 3.1. A regularity stucture is a triple $\mathcal{T} := (A, T, G)$ where :

- $A \subset \mathbb{R}$ is bounded from below and locally finite,
- $T = \bigoplus_{\alpha \in A} T_\alpha$ is a vector space graded by A with each T_α being a Banach space,
- G is a group of continuous operator on T such that for all $\Gamma \in G$, $\alpha \in A$, $\tau \in T_\alpha$, we have $\Gamma\tau - \tau \in \bigoplus_{\beta<\alpha} T_\beta$.

Remarque 3.2. For $\tau \in T$, we denote by $\|\tau\|_\alpha$ the norm of the component of τ in T_α.

Remarque 3.3. If A is infinite, T is the set of formal series $\sum_{\alpha \in A} \tau_\alpha$ where only a finitely many τ_α are non zero.

We need a map to build actual distributions from those algebraic objects, this is the role of the model:

Définition 3.4 (Model)**.** Given a regularity stucture $\mathcal{T}$ and an integer $d \geq 1$, a model for $\mathcal{T}$ on $\mathbb{R}^d$ consists of maps

$$\begin{array}{rcl} \Pi : \mathbb{R}^d & \longrightarrow & \mathcal{L}(T, \mathcal{S}'(\mathbb{R}^d)), \\ x & \longmapsto & \Pi_x, \end{array} \qquad \begin{array}{rcl} \Gamma : \mathbb{R}^d \times \mathbb{R}^d & \longrightarrow & G, \\ (x,y) & \longmapsto & \Gamma_{xy}, \end{array}$$

such that $\forall x, y, z \in \mathbb{R}^d$, $\quad \Gamma_{xy}\Gamma_{yz} = \Gamma_{xz}$ and $\Pi_x \Gamma_{xy} = \Pi_y$, and such that the following estimates hold. Given $r > |\inf A|$, for any compact set $K \subset \mathbb{R}^d$ and constant $\gamma > 0$, there exists a constant C such that :

$$\left|(\Pi_x \tau)(\phi_x^\lambda)\right| \leq C\lambda^{|\tau|} \|\tau\|_\alpha , \quad \|\Gamma_{xy}\tau\|_\beta \leq C\, |x-y|^{\alpha-\beta}\, \|\tau\|_\alpha \tag{3.1}$$

uniformly on $\phi \in \mathcal{B}^r$, $x, y \in K$, $\lambda \in (0,1]$, $\tau \in T_\alpha$ with $\alpha \leq \gamma$, and $\beta < \alpha$.

Définition 3.5. Given a regularity structure $\mathcal{T}$ equipped with a model (Π, Γ) over $\mathbb{R}^d$, we introduce the space of modelled distribution $\mathcal{D}^\gamma(\mathcal{T}, \Gamma)$ as the set of functions $f : \mathbb{R}^d \to \bigoplus_{\alpha<\gamma} T_\alpha$ such that for any compact $K \subset \mathbb{R}^d$ and $\alpha < \gamma$,

$$\|f(x) - \Gamma_{xy} f(y)\|_\alpha \lesssim |x-y|^{\gamma-\alpha} ,$$

uniformly over $x, y \in K$.

Remarque 3.6. Let $f \in \mathcal{D}^\gamma$ be a modelled distribution, and let $\alpha := \inf A$. Then, $(\Pi_x f(x))_{x\in R^d}$ is an (α,γ) coherent germ. As a result, if $\gamma > 0$, we can apply the reconstruction theorem. We can thus define a reconstruction map $\mathcal{R} : (\Pi, \Gamma, f) \mapsto \mathcal{R}(f) := \mathcal{R}((\Pi_x f(x)))$. It turns out to be locally Lipschitz when the distance between (Π,Γ,f) and $(\bar\Pi,\bar\Gamma,\bar f)$ is given by the smallest constant κ such that :

$$\begin{aligned}\left\|f(x) - \bar f(x) - \Gamma_{xy} f(y) - \bar\Gamma_{xy}\bar f(y)\right\|_\alpha &\le \kappa\,|x-y|^{\gamma-\alpha},\\ \left|(\Pi_x\tau - \bar\Pi_x\tau)(\phi_x^\lambda)\right| &\le \kappa\lambda^\alpha\,\|\tau\|,\\ \left\|\Gamma_{xy}\tau - \bar\Gamma_{xy}\tau\right\|_\beta &\le \kappa\,|x-y|^{\alpha-\beta}\,\|\tau\|.\end{aligned}$$

uniformly locally on x, y, $\phi \in \mathcal{B}^r$, $\tau \in T$, $\beta < \alpha \le \gamma$.

Proof. Let us prove that $(\Pi_x f(x))_{x\in R^d}$ is an (α,γ)-coherent germ. Let $K \subset \mathbb{R}^d$ a compact set and $x, y \in K$. For all $\beta \in A$, $\beta < \gamma$, denote $\delta_\beta^{x,y}$ the projection of $f(x) - \Gamma_{xy} f(y)$ on T_β. We have :

$$\begin{aligned}\left|(\Pi_x f(x) - \Pi_y f(y))(\phi_x^\lambda)\right| &= \left|(\Pi_x(f(x) - \Gamma_{xy}f(y)))(\phi_x^\lambda)\right|\\ &\lesssim \sum_{\substack{\beta\in A\\ \alpha\le\beta<\gamma}} \left|(\Pi_x(\delta_\beta^{x,y}))(\phi_x^\lambda)\right|\\ &\lesssim \sum_{\substack{\beta\in A\\ \alpha\le\beta<\gamma}} \lambda^\beta\,\|f(x) - \Gamma_{xy}f(y)\|_\beta \quad \text{(by definition of a model)}\\ &\lesssim \sum_{\substack{\beta\in A\\ \alpha\le\beta<\gamma}} \lambda^\beta\,|x-y|^{\gamma-\beta} \quad (f\in\mathcal{D}^\gamma)\\ &\lesssim \lambda^\alpha \max_{\substack{\beta\in A\\ \alpha\le\beta<\gamma}} \lambda^{\beta-\alpha}\,|x-y|^{\gamma-\alpha-(\beta-\alpha)}\\ &\lesssim \lambda^\alpha(\lambda + |x-y|)^{\gamma-\alpha}.\end{aligned}$$

□

The space of modelled distribution is central because it is the space in which we are able to state our solution problem as a fixed-point problem. To properly define the associated contraction map, we need two tools : a reconstruction theorem, and multilevel Schauder-estimates. The reconstruction theorem is direct corollary of theorem 2.17 and remark 3.6.

Corollaire 3.7 (Reconstruction theorem for modelled distribution)**.** *Let $\mathcal{T}$ be a regularity structure equipped with a model (Π,Γ) over $\mathbb{R}^d$. Let $\gamma > 0$ and $f \in \mathcal{D}^\gamma(\mathcal{T},\Gamma)$ be a modelled distribution. Finally, let $\alpha := \inf A$ and fix $r > |\alpha|$. Then there exists a reconstruction map $\mathcal{R} : \mathcal{D}^\gamma(\mathcal{T},\Gamma) \to \mathcal{C}^\alpha$ such that for every compact set $K \subset \mathbb{R}^n$,*

$$|(\mathcal{R}(f) - \Pi_x f(x))(\phi_x)| \lesssim \lambda^\gamma$$

Chapter 4

The ϕ^4_3 model

4.1 Introduction

In this chapter, we focus on presenting the geometric and diagram-free approach of article [3], which was the main topic of the internship. Recall the ϕ_3^4 equation :

$$L\phi - \lambda\phi^3 = \xi, \tag{4.1}$$

where $L = \partial_0 - \sum_{i=1}^d \partial_i^2$ is the heat operator in $d = 3$ space dimensions, and ξ represents a space-time white noise.

As discussed in Chapter 1, the classical Da Prato-Debussche approach, while highly effective for the ϕ_2^4 model, fails in $d = 3$ [1]. Indeed, in three spatial dimensions, the linear stochastic convolution Π_0 is a genuine distribution of local Hölder regularity $-\frac{1}{2}-$. As a result, the non-linear remainder $\psi = \phi - \Pi_0$ does not possess enough regularity to make sense of the products $\psi\Pi_0^2$ or $\psi^2\Pi_0$ via classical Young multiplication. To resolve this, we draw on the theory of regularity structures, in a diagram-free approach pioneered by Linares, Otto, Tempelmayr, and Tsatsoulis, which indexes the model components by multi-indices β rather than trees.

The stochastic estimates of the model are obtained by leveraging a *Spectral Gap Inequality* on the noise ensemble, which controls the probabilistic L^p-norms of the model by means of their directional *Malliavin derivatives* $\delta\Pi$. This reduces the probabilistic problem of model bounds to the analytic task of reconstructing and integrating the Malliavin increments, which are governed by a robust relation $\delta\Pi \to \delta\Pi^-$ that does not depend on the diverging counterterms as shown in section 4.3.

We consider the space-time noise ξ as a random Schwartz distribution on the d-dimensional space-time. More precisely, ξ is a centered Gaussian process defined on the space of Schwartz functions, fully characterized by its covariance structure. For any test functions ζ and ζ', the covariance of the dual pairing satisfies the relation:

$$\mathbb{E}[(\xi,\zeta)(\xi,\zeta')] = \int_{\mathbb{R}^{1+d}} \zeta(x)\zeta'(x)\,dx. \tag{4.2}$$

This spatial inner product makes the law of the noise invariant under space-time translations, spatial reflections, and parity. Moreover, the noise possesses a parabolic scale invariance in law under the scaling operator R_r, which scales the time-like variable x_0 by r^2 and the space-like variables x_i by r. The scaling operator is defined for any scale $r > 0$ by:

$$R_r x = (r^2 x_0, r x_1, \dots, r x_d). \tag{4.3}$$

This scale invariance is written as:

$$\xi(R_r\cdot) \stackrel{\text{law}}{=} r^{s-\frac{D}{2}}\xi, \tag{4.4}$$

where s is a real exponent and $D = d + 2$ represents the effective parabolic dimension of the space-time.

To analyze the behavior of the system under scaling, we consider a blow-up or zoom-in transformation of the noise and the solution. Under this scaling, the linear solution ϕ_0 of the equation $L\phi_0 = \xi$ has a regularity exponent $\alpha = 2 + s - D/2$. The original semi-linear equation is invariant under this rescaling provided that we adjust the strength of the cubic non-linearity by a factor $r^{2(1+\alpha)}$. This leads to the definition of the subcritical regime, which is characterized by the condition:

$$\alpha + 1 > 0. \tag{4.5}$$

In this regime, the exponent associated with the scaled coupling constant is strictly positive, which means that the effect of the non-linearity vanishes on small scales. This ensures that the equation is super-renormalizable and can be resolved locally using a finite number of counterterms.

To regularize the singular equation, we introduce a mollified noise at scale ρ and modify the partial differential equation by introducing a deterministic, regularization-dependent counterterm. Symmetries under space-time translations, state reflection parity, and spatial reflections significantly restrict the allowed counterterms. First, stationarity under space-time translations dictates that the counterterm coefficients must be space-time constants. Second, because the noise is a centered Gaussian, its law is symmetric under state reflection, meaning $-\xi \stackrel{\text{law}}{=} \xi$. Since the cubic non-linearity is also odd, this state reflection symmetry $-\phi \stackrel{\text{law}}{=} \phi$ propagates through the multi-indices. On the level of the model components, the noise homogeneity which keeps track of the number of noises in the multi-index β dictates that:

$$(-1)^{1+\sum_n \beta(n)}\Pi_\beta \stackrel{\text{law}}{=} \Pi_\beta \quad \text{and} \quad (-1)^{1+\sum_n \beta(n)}\Pi^-_\beta \stackrel{\text{law}}{=} \Pi^-_\beta. \tag{4.6}$$

This implies that the counterterm coefficients must vanish unless the exponent $1+\sum_n \beta(n)$ is even, which rules out terms like $h(\rho)1$ or $h(\rho)\phi^2$. Third, spatial reflection parity across the $\{x_i = 0\}$ plane dictates that:

$$(-1)^{\sum_n n_i\beta(n)}\Pi_\beta(R_i\cdot) \stackrel{\text{law}}{=} \Pi_\beta \quad \text{and} \quad (-1)^{\sum_n n_i\beta(n)}\Pi^-_\beta(R_i\cdot) \stackrel{\text{law}}{=} \Pi^-_\beta. \tag{4.7}$$

Evaluating at the base point $x = 0$ against a spatially symmetric kernel, this forces the expectation to vanish if the spatial parity exponent is odd, ruling out gradient terms of the form $h_i(\rho)\partial_i\phi$. Combining these constraints with the subcriticality condition that only permits counterterms associated with multi-indices of negative homogeneity ($|\beta| < 2$), we reduce the allowed counterterms to the linear form $h = h(\rho)\phi$. The coefficient $h(\rho)$ is represented as a formal power series in the coupling constant λ given by:

$$h(\rho) = \sum_{k\geq 0} c_k\lambda^k. \tag{4.8}$$

The constant coefficients c_k are deterministic and vanish unless k is strictly positive and satisfies the scaling restriction:

$$0 < k < (1+\alpha)^{-1}. \tag{4.9}$$

We parameterize the non-linear solution manifold using a coordinate system on the parameter space of the coupling constant and the space-time analytic functions. Let p be an analytic function representing the solution in the linear case. We introduce coordinates $z_n[p] = \frac{1}{n!}\partial^n p(0)$ for space-time multi-indices $n = (n_0, \ldots, n_d)$, and $z_3[\lambda] = \lambda$ for the coupling constant. For any multi-index β over the index set, we define the coordinate monomial:

$$z^\beta = z_3^{\beta(3)} \prod_n z_n^{\beta(n)}. \tag{4.10}$$

We then make an ansatz for the solution ϕ as a formal power series given by:

$$\phi = \sum_\beta z^\beta \Pi_\beta, \tag{4.11}$$

where the coefficients Π_β are random space-time functions. To ensure consistency for the linear case when the coupling constant vanishes, the polynomial boundary conditions are uniquely fixed on the components of zero coupling homogeneity. That is, for any multi-index β such that $\beta(3) = 0$ but β is not the zero multi-index, we have:

$$\Pi_\beta(y) = \begin{cases} y^n & \text{if } \beta = \delta_n, \\ 0 & \text{otherwise,} \end{cases} \tag{4.12}$$

where δ_n represents the multi-index that assigns the value one to the coordinate index n and zero to all other entries. This represents a compact separation of variables between the parameter space coordinates and the random space-time fluctuations of the model.

We introduce the formal power series Π^- to represent the right-hand side of the renormalized equation, including the non-linear cubic term and the counterterm. The component-wise description of the system is captured by the relation:

$$\Pi^- = z_3 \Pi^3 + c\Pi + \xi_\rho 1, \tag{4.13}$$

where c represents the formal power series of the renormalization constants and the product is defined via the algebraic multiplication of formal power series. Component-wise, the elements of the right-hand side are determined by the formula:

$$\Pi^-_\beta = \sum_{\beta_1+\beta_2+\beta_3+\delta_3=\beta} \Pi_{\beta_1}\Pi_{\beta_2}\Pi_{\beta_3} + \sum_{\beta_1+\beta_2=\beta,\, \beta_1(3)>0,\, \forall n: \beta_1(n)=0} c_{\beta_1}\Pi_{\beta_2} + \xi_\rho \delta_{0\beta}. \tag{4.14}$$

With these notations, the renormalized partial differential equation can be compactly written as:

$$L\Pi - \Pi^- = 0 \quad \text{mod analytic functions.} \tag{4.15}$$

This relation holds component-wise for each multi-index β, providing a strictly triangular hierarchy of equations that allows for the inductive construction of the model components.

Since the parameterization of the solution manifold depends on the choice of the origin, we introduce a general base-point x in space-time to define a translation-invariant model. This change of base-point leads to two different parameterizations of the same solution, which are related by a random parameter transformation. This transformation induces, by pull-back, a linear endomorphism of the space of formal power series denoted by Γ_x^*. Algebraically, the transition map Γ_x^* is a multiplicative algebra automorphism. This means that for any power series π and π', it satisfies the relation:

$$\Gamma_x^*(\pi\pi') = (\Gamma_x^*\pi)(\Gamma_x^*\pi') \tag{4.16}$$

and maps the unit element to itself. Furthermore, it leaves the coupling constant invariant, meaning:

$$\Gamma_x^* z_3 = z_3, \tag{4.17}$$

and consequently acts trivially on the renormalization constants:

$$\Gamma_x^* c = c. \tag{4.18}$$

The shift of the base-point from the origin to x is then algebraically represented by the identities:

$$\Pi = \Gamma_x^* \Pi_x, \tag{4.19}$$

$$\Pi^- = \Gamma_x^* \Pi_x^-. \tag{4.20}$$

4.2 The spectral gap inequality

Structurally, the spectral inequality inequality behaves like an infinite-dimensional Poincaré inequality on the space of space-time fields. It is specified over a translation-invariant Hilbert space (typically an L^2-based fractional Sobolev or Besov space) that plays the role of the Cameron-Martin space in Gaussian measures. For a generic integrable random variable $F = F[\xi]$, the inequality bounds its variance by the expectation of the norm of its Malliavin derivative:

$$\mathbb{E}(F - \mathbb{E}F)^2 \leq \mathbb{E} \left\| \frac{\partial F}{\partial \xi} \right\|_{\dot{H}^{-s}}^2 \tag{4.21}$$

While the BPHZ choice of counterterms is designed to annihilate the expectation of the random variables of negative homogeneity ($\mathbb{E}\Pi^- = 0$), the spectral gap inequality controls their variance. This effectively reduces the stochastic task of estimating the L^p moments of the model to the analytical task of controlling its Malliavin derivative. Furthermore, this Poincaré-type bound can be upgraded via Leibniz's rule and Hölder's inequality to an L^p version in probability for any $2 \leq p < \infty$, bounding the L^p fluctuations of F by the L^p norm of its Malliavin derivative :

$$\mathbb{E}^{\frac{1}{p}} \left| F - \mathbb{E}F \right|^p \lesssim \mathbb{E}^{\frac{1}{p}} \left\| \frac{\partial F}{\partial \xi} \right\|_{\dot{H}^{-s}}^p$$

Furthermore, for a centered Gaussian ensemble that satisfies $\mathrm{Var}(\xi, \zeta) \leq \|\zeta\|_{\dot{H}^{-s}}^2$, the variance of a random variable F is dominated by the expectation of the carré-du-champs of its Malliavin derivative :

$$\left\| \frac{\partial F}{\partial \xi} \right\|_{\dot{H}^{-s}}^2 = \left(\sup_{\partial \xi} \frac{\partial F}{\|\partial \xi\|_{\dot{H}^s}} \right)^2.$$

So we can reformulate :

$$\mathbb{E}^{\frac{1}{p}} \left| F - \mathbb{E}F \right|^p \lesssim \sup\{\mathbb{E}\partial F | \delta\xi \text{ r.v. with } \mathbb{E} \|\delta\xi\|_{\dot{H}^s}^{p^*} \leq 1\},$$

where p^* is the dual exponent of p. We will use it to estimate the random variable $F = \Pi_{\beta r}^-(0)$, in the form :

$$\mathbb{E}^{\frac{1}{p}} \left| \Pi_{\beta r}^-(0) \right|^p \lesssim \left| \mathbb{E}\Pi_{\beta r}^-(0) \right| + \sup\{\mathbb{E}^{\frac{1}{q}} \left| \delta\Pi_{\beta r}^-(0) \right|^q | \delta\xi \text{ r.v. with } \mathbb{E} \|\delta\xi\|_{\dot{H}^s}^{p^*} \leq 1\}.$$

The argument we will use is based on induction. We will simultaneously prove estimates for all p, q in the range $1 \le q < p^* \le 2 \le p$, so that it will not be a problem to appeal recursively to these estimates with different exponents q', and p'. In practice, in the induction, we fix p, q and $\delta\xi \in \dot{H}^s$ with $\mathbb{E}\,\|\delta\xi\|_{\dot{H}^s}^{p^*} \le 1$.

This formulation enables one to capture the gain in regularity of the model under perturbation by describing the directional Malliavin derivative as a modeled distribution, thereby linking the stochastic estimates directly to pathwise analytical tools such as the reconstruction and integration theorems

4.3 A robust relation $\delta\Pi \longrightarrow \delta\Pi^-$

We established that :

$$\partial^{\mathbf{n}}(\delta\Pi - d\Gamma_x^*\Pi_x)(x) = 0 \quad \text{for } |\mathbf{n}| < 2+s. \tag{4.22}$$

We will see how this specifies uniquely the coefficients $\mathrm{d}\Gamma_x^*$. Indeed, by definition,

$$\mathrm{d}\Gamma_x^* := \sum_{|\mathbf{n}|<2+s} \mathrm{d}\pi_x^{(n)}\Gamma_x^*\partial_{z_\mathbf{n}}.$$

And as

$$\partial_{z_\mathbf{n}} z_\mathbf{m} = \delta_{\mathbf{n},\mathbf{m}},$$

It follows that

$$\forall \mathbf{m}, |\mathbf{m}| \geq 2+s, \mathrm{d}\Gamma_x^* z_m = 0.$$

Additionnaly, $\mathrm{d}\Gamma_x^*$ behaves like a derivation :

$$\begin{cases} \mathrm{d}\Gamma_x^*\pi\pi' = (\mathrm{d}\Gamma_x^*\pi)(\Gamma_x^*\pi') + (\Gamma_x^*\pi)(\mathrm{d}\Gamma_x^*\pi'), \\ \mathrm{d}\Gamma_x^* 1 = 0, \\ \mathrm{d}\Gamma_x^*\mathbf{z}_3 = 0. \end{cases}$$

This follows directly from the behaviour of $\partial_{\mathbf{z}_n}$ which is a derivation, and Γ_x^* which is an algebra morphism. As a consequence, together with its finiteness property, $\mathrm{d}\Gamma_x^*$ is determined by its values on the finitely many coordinates $\{\mathbf{z}_n\}_{|\mathbf{n}|<2+s}$.

Knowing $\delta\Pi$ and (4.22), we actually know $\mathrm{d}\Gamma_x^*\partial^{\mathbf{n}}\Pi_x(x)$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2+s$. We also know this fact about the jet:

$$\frac{1}{\mathbf{n}!}\partial^{\mathbf{n}}\Pi_x(x) - \mathbf{z}_\mathbf{n} \in \mathbb{R}[[\mathbf{z}_3, \{\mathbf{z}_\mathbf{m}\}_{|\mathbf{m}|<|\mathbf{n}|}]]. \tag{4.23}$$

Now, an induction argument shows that we thus know $\mathrm{d}\Gamma_x^*\mathbf{z}_n$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2+s$.

Proof. The base case is is clear because $\Pi_x(x) - \mathbf{z}_\mathbf{0} \in \mathbb{R}[[\mathbf{z}_3]]$ and $\mathrm{d}\Gamma_x^*\mathbf{z}_3 = 0$ so

$$\mathrm{d}\Gamma_x^*\mathbf{z}_\mathbf{0} = \mathrm{d}\Gamma_x^*\Pi_x(x).$$

Now let $k \in \mathbb{N}$, $k < 2+s$ be such that we know $\mathrm{d}\Gamma_x^*\mathbf{z}_\mathbf{m}$ for all $\mathbf{m}$ such that $|\mathbf{m}| < k$. For all $\mathbf{n}$ with $|\mathbf{n}| = k$, (4.23) shows that $\mathrm{d}\Gamma_x^*\mathbf{z}_\mathbf{n}$ is a function of $\mathrm{d}\Gamma_x^*\partial^{\mathbf{n}}\Pi_x(x)$ and some $\mathrm{d}\Gamma_x^*\mathbf{z}_\mathbf{m}$ with $|\mathbf{m}| < k$, which are known by hypothesis.

□

This will be useful to explicit a robust relation $\delta\Pi \longrightarrow \delta\Pi^-$. By robust we mean that the map doesn't rely on diverging constant such as c, like if we used the first definition of $\delta\Pi^-$:

$$\delta\Pi^- := (3\mathbf{z}_3\Pi^2 + c)\delta\Pi + \delta\xi_\rho\mathbf{1}.$$

We claim that :

$$(\delta\Pi^- - \delta\xi_\rho\mathbf{1} - \mathrm{d}\Gamma_x^*\Pi_x^-)(x) = 0 \tag{4.24}$$

Proof. Let's work out $\mathrm{d}\Gamma_x^*\Pi_x^-(x)$:

$$\begin{aligned}
\mathrm{d}\Gamma_x^*\Pi_x^-(x) &= \mathrm{d}\Gamma_x^*(\mathbf{z}_3\Pi_x^3 + c\Pi_x + \xi_\rho\mathbf{1})(x) \\
&= ((\Gamma_x^*\mathbf{z}_3)(\mathrm{d}\Gamma_x^*\Pi_x^3) + (\Gamma_x^*c)(\mathrm{d}\Gamma_x^*\Pi_x) + \mathrm{d}\Gamma_x^*\xi_\rho\mathbf{1})(x) \quad \text{(derivation properties)} \\
&= (3\mathbf{z}_3(\mathrm{d}\Gamma_x^*\Pi_x)(\Gamma_x^*\Pi_x^2) + c\,\mathrm{d}\Gamma_x^*\Pi_x)(x) \quad \text{(same properties, and } \Gamma_x^*c = c, \Gamma_x^*\mathbf{z}_3 = \mathbf{z}_3\text{)} \\
&= (3\mathbf{z}_3\Pi^2 + c)(\mathrm{d}\Gamma_x^*\Pi_x)(x) \quad \text{(because } \Gamma_x^*\Pi_x = \Pi\text{)} \\
&= (3\mathbf{z}_3\Pi^2 + c)(\delta\Pi)(x) \quad \text{(by (4.23) for } \mathbf{n} = \mathbf{0}\text{)} \\
&= (\delta\Pi^- - \delta\xi_\rho)(x) \quad \text{(by definition of } \delta\Pi^-\text{)}
\end{aligned}$$

□

The calculation is easier in this direction altough we are truely going in the opposite direction. The success of our approach may be due to the fact that we use an approximation of $\delta\Pi$ at each x with the germ $\mathrm{d}\Gamma_x^*\Pi_x$, as opposed to only utilizing $\Pi_0; \Pi_0^-$.

Equation (4.24) will give us the qualitative result we need to quantitatively reconstruct $\delta\Pi^-$ from the germ $\mathrm{d}\Gamma_x^*\Pi_x^-$ (see section 3.2 of [**BOT_2024**]). We also see the main induction argument coming : from $\delta\Pi$ we get this relation, and we will integrate the estimate we obtain on $\delta\Pi^-$ to get an estimate on $\delta\Pi$.

4.4 Main result

The goal is to prove convergence of the model in the forme of the following estimates :

$$\mathbb{E}^{\frac{1}{p}}\left|\Pi_{x\beta r}(x)\right|^p \lesssim r^{|\beta|}, \tag{4.25}$$
$$\mathbb{E}^{\frac{1}{p}}\left|\Pi^-_{x\beta r}(x)\right|^p \lesssim r^{|\beta|-2}, \tag{4.26}$$
$$\mathbb{E}^{\frac{1}{p}}\left|(\Gamma^*_x)^\gamma_\beta\right|^p \lesssim r^{|\beta|-|\gamma|} \quad \text{for all populated } \gamma. \tag{4.27}$$

The first two estimates are homogeneity estimates on the germs Π_x and Π^-_x, and together with the last one, we also get a coherence estimate on these two germs.

$$\mathbb{E}^{\frac{1}{p}}\left|(\Pi_{x+y\beta}-\Pi_{x\beta})_r(x)\right|^p \lesssim r^\alpha(r+|y|)^{|\beta|-\alpha}, \tag{4.28}$$
$$\mathbb{E}^{\frac{1}{p}}\left|\Pi^-_{\beta r}(x)\right|^p \lesssim r^{3\alpha}(r+|x|)^{|\beta|-2-3\alpha}, \quad \text{provided } \beta\neq 0. \tag{4.29}$$

This estimate gives us useful insight : the increments $\Pi_{x+y\beta}-\Pi_{x\beta}$ have local regularity α (which is negative) at an arbitrary point, but vanishes at the order $|\beta|$ when y goes to 0. Finally, at infinity, they grow at order $|\beta|-\alpha$.

Proof. Proof of (4.28) We know that $\Pi_{x+y\beta}-\Pi_{x\beta} = (\Gamma^*_y - id)\Pi_{x\beta} = \sum_\gamma (\Gamma^*_y-id)^\gamma_\beta \Pi_{x\gamma}$. We apply the mollification operator $(\cdot)_r$ and evaluate at x. Using the triangle inequality with respect to the norm $\mathbb{E}^{\frac{1}{p}}|\cdot|^p$, and then Hölder's inequality, we get :

$$\mathbb{E}^{\frac{1}{p}}\left|(\Pi_{x+y\beta}-\Pi_{x\beta})_r(x)\right|^p \lesssim \sum_\gamma \mathbb{E}^{\frac{1}{2p}}\left|(\Gamma^*_x-id)^\beta_\gamma\right|^{2p}\mathbb{E}^{\frac{1}{2p}}\left|\Pi_{x\gamma r}(x)\right|^{2p}. \tag{4.30}$$

Now appealing to $|\gamma|\geq|\beta| \implies (\Gamma^*_x-id)^\gamma_\beta = 0$ to restrict the sum and get rid of id, and estimates (4.25) and (4.27), we get :

$$\mathbb{E}^{\frac{1}{p}}\left|(\Pi_{x+y\beta}-\Pi_{x\beta})_r(x)\right|^p \lesssim \sum_{|\gamma|\leq|\beta|} |y|^{|\beta|-|\gamma|} r^{|\gamma|}. \tag{4.31}$$

We also know that $|\gamma|\geq\alpha$ for all populated γ (see (62) in [3], α is the roughest regularity of the model, it is the regularity of Π_0), so we can restrict the sum to $|\gamma|\geq\alpha$. Factorizing by r^α, and using a binomial expansion, we get (4.28). $\square$

The proof for Π^- is analog.

We are not trying to apply a reconstruction theorem here, but coherence and homogeneity are the two relevant metrics to measure germs of distributions. In order to prove these estimates, we will first prove similar estimates on $\delta\Pi$ and $\delta\Pi^-$, which are the Malliavin derivatives of Π and Π^- with respect to the noise ξ, or rather on their germs $\mathrm{d}\Gamma^*_x\Pi_x$ and $\mathrm{d}\Gamma^*_x\Pi^-_x$ (cf section 2.6 from [3] to see why these are the right germs). We will take advantage of the fact that $\delta\Pi$ and $\delta\Pi^-$ are related by a robust relation, and that they are more regular, by a order $\frac{D}{2}$.

The shape of the estimate will be slighty different as we are going to take a local L^2-norm in x instead of an L^∞ norm. This is dictated by what we are able to prove for the base case : a pointwise estimate is to strong, and we use an L^2-norm because of the L^2 nature of $\dot{H}_s$ ($\ni \delta\Pi^-_0 = \delta\xi_\rho$). The proof of the base case relies on Fourier transform. Finally, we cannot expect square integrability of $\mathbb{E}^{\frac{2}{q}}\left|(\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_{\beta r}(x)\right|^q$ at infinity, so we take a local norm.

We will thus try to prove the following estimates :

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|(d\Gamma^*_{x+y}\Pi^-_{x+y} - d\Gamma^*_x\Pi^-_x)_{\beta r}(x+y)\right|^q)^{\frac{1}{2}} \\ \lesssim r^{2\alpha}(r+|y|)^{\alpha+\frac{D}{2}}(|y|+r+R)^{|\beta|-2-3\alpha} \quad \text{(coherence)} \quad (4.32)$$

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|(d\Gamma^*_x)^\gamma_\beta\right|^q)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+\beta-\gamma} \quad \text{for populated } \gamma \neq \text{pp} \quad \text{(homogeneity)} \quad (4.33)$$

This will allow us to prove by reconstruction that :

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^- - d\Gamma^*_x\Pi^-_x)_{\beta r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{3\alpha+\frac{D}{2}}(r+R)^{|\beta|-2-3\alpha} \quad \text{(reconstruction)} \quad (4.34)$$

And deduce a reconstruction bound for $\delta\Pi$ by integration (see below) :

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} |(\delta\Pi - d\Gamma^*_x\Pi_x)_{\beta r}(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(reconstruction)} \quad (4.35)$$

What is cructial here is the exponent $3\alpha + \frac{D}{2}$, in the coherence estimate of $d\Gamma^*_x\Pi^-_x$ and in the reconstruction bound for $\delta\Pi^-$. It indeed plays the role of γ in the reconstruction theorem presented in [2], and is positive by assumption in [3]. Because of this, the reconstuction theorem ensures uniqueness of the reconstruction $\delta\Pi^-$. The same goes for $\delta\Pi$ with the exponent $\alpha + \frac{D}{2}$ which is even more positive as $\alpha < 0$. This cascades down to show that our model is uniquely caracterized.

Actually we can't directly integrate (4.34) because of terms of to high homogeneity in $d\Gamma^*_x\Pi^-_x$. We will need to split this sum into two parts :

$$(d\Gamma^*_x\Pi^-_x)_\beta = \sum_{|\gamma|\geq 2+s}(d\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma} + \sum_{|\gamma|<2+s}(d\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma}$$

The first part is the one we will be able to control by homogeneity, because of the lower bound on $|\gamma|$:

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|\sum_{|\gamma|\geq 2+s}(d\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(homogeneity)} \quad (4.36)$$

we thus get by triangle inequality and (4.34):

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^-_\beta - \sum_{|\gamma|<2+s}(d\Gamma^*_x)^\gamma_\beta\Pi^-_{x\gamma})_r(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-\alpha}, \quad (4.37)$$

which we are able to integrateto find :

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi_\beta - \sum_{|\gamma|<2+s}(d\Gamma^*_x)^\gamma_\beta\Pi_{x\gamma})_r(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha}. \quad (4.38)$$

Because we also have the homogeneity estimate of the high homogeneity part of $d\Gamma^*_x\Pi_x$:

$$(\int_{B_R} dx\mathbb{E}^{\frac{2}{q}} \left|\sum_{|\gamma|\geq 2+s}(d\Gamma^*_x)^\gamma_\beta\Pi_{x\gamma r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(homogeneity)}, \quad (4.39)$$

we are able to prove (4.35) by triangle inequality.

Let us prove the homogeneity estimate (4.36), but first try the direct approach to prove the homogeneity bound of the germ $\mathrm{d}\Gamma_x^* \delta\Pi_x^-$ without splitting, to see the difference in exponents. We will end up using both computation anyway.

$$
\begin{aligned}
\Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \big| (\mathrm{d}\Gamma_x^* \Pi_x^-)_\beta \big|^q \Big)^{\frac12} &= \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \Big| \sum_\gamma (\mathrm{d}\Gamma_x^*)_\beta^\gamma \, \Pi_{x\gamma r}^-(x) \Big|^q \Big)^{\frac12} \\
&\lesssim \sum_\gamma \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{2q}} \big| (\mathrm{d}\Gamma_x^*)_\beta^\gamma \big|^{2q} \mathbb{E}^{\frac{2}{2q}} \big| \Pi_{x\gamma r}^-(x) \big|^{2q} \Big)^{\frac12} \quad \text{(Hölder)} \\
&\lesssim \sum_{\substack{\gamma \neq \mathrm{pp} \\ |\gamma|_\prec < |\beta|_\prec}} \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{2q}} \big| (\mathrm{d}\Gamma_x^*)_\beta^\gamma \big|^{2q} \Big)^{\frac12} \sup_{x \in B_R} \mathbb{E}^{\frac{1}{2q}} \big| \Pi_{x\gamma r}^-(x) \big|^{2q} \\
&\lesssim \sum_{\alpha \leq |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - 2} \quad \text{(by induction hypothesis)} \\
&\lesssim r^{\alpha - 2} \sum_{\alpha \leq |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - \alpha} \\
&\lesssim r^{\alpha - 2} (R + r)^{\frac{D}{2} + |\beta| - \alpha}
\end{aligned}
$$

The condition $\gamma \neq \mathrm{pp}$ comes from the fact that $\gamma = \mathrm{pp} \implies \Pi_{x\gamma}^- = 0$. The condition $|\gamma|_\prec < |\beta|_\prec$ comes from the stric triangularity of $\mathrm{d}\Gamma_x^*$ with respect to $|\cdot|_\prec$. Finally, we always have $|\gamma| \geq \alpha$, and $|\gamma| < |\beta| + \frac{D}{2}$ also comes from population conditions on $\mathrm{d}\Gamma_x^*$.

Now, say that we truncate the sum from below by $|\gamma| \geq 2 + s = \alpha + \frac{D}{2}$, and try to prove (4.36) :

$$
\begin{aligned}
\Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \Big| \sum_{|\gamma| \geq 2 + s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \, \Pi_{x\gamma r}^-(x) \Big|^q \Big)^{\frac12} &\lesssim \sum_{\alpha + \frac{D}{2} \leq |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - 2} \\
&\lesssim r^{\alpha + \frac{D}{2} - 2} \sum_{\alpha + \frac{D}{2} \leq |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - \alpha - \frac{D}{2}} \\
&\lesssim r^{\alpha + \frac{D}{2} - 2} (R + r)^{|\beta| - \alpha}
\end{aligned}
$$

which is better, and matches the reconstruction bound (4.34). So this allow us to not degrade our exponents when using the triangle inequality to get (4.37), which in turn allows us to get the good reconstruction bound on $\delta\Pi$ (4.35).

4.5 Reconstruction of $\mathrm{d}\Gamma_x^*\Pi_x^-$

Proof of (4.34). For convenience, denote $F_x \coloneqq (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta$ and introduce the diagonal evaluation for this germ : $(EF)(x) = F_x(x)$. We are going to prove,

$$\Big(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(EF_\tau - F_{x\tau})_{T-\tau}(x)|^q\Big)^{\frac{1}{2}} \lesssim (\sqrt[4]{T})^{3\alpha+\frac{D}{2}}(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha}, \tag{4.40}$$

because from (4.24), we have the qualitative : $\delta\Pi_\beta^-(x) = \lim_{\tau\to 0}(\mathrm{d}\Gamma_x^*\Pi_x^-)_{\beta\tau}(x) = \lim_{\tau\to 0} EF_\tau(x)$, so taking the limit $\tau \to 0$, we hope to recover the targeted estimate.

Let $n \in \mathbb{N}$ and $\tau = 2^{-n}T$. Then, we can write

$$\begin{aligned}
(EF_\tau - F_{x\tau})_{T-\tau}(x) &= (EF_\tau)_{T-\tau} - (EF_{2\tau})_{T-2\tau} \\
&+ (EF_{2\tau})_{T-2\tau} - (EF_{4\tau})_{T-2\tau} \\
&+ (EF_{4\tau})_{T-4\tau} - (EF_{8\tau})_{T-8\tau} + \dots \\
&+ (EF_{\frac{T}{2}})_{\frac{T}{2}} - EF_T(x) \\
&= \sum_{k=0}^{n-1}(EF_{2^{k-n}T})_{(1-2^{k-n})T} - (EF_{2^{k+1-n}T})_{(1-2^{k+1-n})T}
\end{aligned}$$

Let $t = 2^{k-n}T$, we then have to estimate $(EF_t)_{T-t} - (EF_{2t})_{T-2t} = ((EF_t)_t - EF_{2t})_{T-2t}$ (semi-group property).

$$\begin{aligned}
((EF_t)_t - EF_{2t})_{T-2t}(x) &= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t)_t - EF_{2t})(x-y) \\
&= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t)_t - F_{x-y2t})(x-y) \\
&= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t) - F_{x-yt})_t(x-y) \\
&= \int \mathrm{d}y\, \psi_{T-2t}(y)\int \mathrm{d}y'\psi_t(y')((EF_t) - F_{x-yt})(x-y-y') \\
&= \int \mathrm{d}y\, \psi_{T-2t}(y)\int \mathrm{d}y'\psi_t(y')(F_{x-y-y't} - F_{x-yt})(x-y-y')
\end{aligned}$$

We see the increments of the germ $\mathrm{d}\Gamma_x^*\Pi_x^-$ appearing. Using the fact that $\int_{B_R}\mathrm{d}x f^2(x+y) \leq \int_{B_{R+|y|}}\mathrm{d}x' f^2(x')$, we have

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(F_{x-y-y't} - F_{x-yt})(x-y-y')|^q)^{\frac{1}{2}} \lesssim (\int_{B_{R+|y|}} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(F_{x-y't} - F_{xt})(x-y')|^q)^{\frac{1}{2}} \tag{4.41}$$

And the r.h.s is estimated by (4.32) with $(\sqrt[4]{t}, -y', R+|y|)$ playing the role of (r, y, R).

Thus, by triangle inequality, we have :

$$\begin{aligned}
&(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |((EF_t)_t - EF_{2t})_{T-2t}(x)|^q)^{\frac{1}{2}} \\
&\lesssim (\sqrt[4]{t})^{2\alpha}\int \mathrm{d}y\, |\psi_{T-2t}(y)|\int \mathrm{d}y'\, |\psi_t(y')|\,(|y'|+\sqrt[4]{t})^{\alpha+\frac{D}{2}}(|y'|+\sqrt[4]{t}+R+|y|)^{|\beta|-2-3\alpha}
\end{aligned} \tag{4.42}$$

Let $\mu = \sqrt[4]{t}$ and $\lambda = \sqrt[4]{T-2t}$. By change of variable,

$$\begin{aligned}(\sqrt[4]{t})^{2\alpha} \int dy\, |\Psi_{T-2t}(y)| \int dy'\, |\Psi_t(y')|\, (|y'| + \sqrt[4]{t})^{\alpha+\frac{D}{2}} (|y'| + \sqrt[4]{t} + R + |y|)^{|\beta|-2-3\alpha} \\ = (\sqrt[4]{t})^{2\alpha} \iint |\Psi_1(z)|\, |\Psi_1(z')| \left[\mu^{\alpha+\frac{D}{2}} (1+|z'|)^{\alpha+\frac{D}{2}}\right] (\mu|z'| + \lambda|z| + \mu + R)^{|\beta|-2-3\alpha}\, dz\, dz'\end{aligned} \tag{4.43}$$

We separate scale variables from integration variables :

$$\mu|z'| + \lambda|z| + \mu + R \leq (\mu + \lambda + R)(1 + |z| + |z'|) \tag{4.44}$$

We obtain :

$$\begin{aligned}\text{l. h. s. of (4.43)} &\leq (\sqrt[4]{t})^{2\alpha} \cdot \mu^{\alpha+\frac{D}{2}} \cdot (\mu + \lambda + R)^{|\beta|-2-3\alpha} \\ &\quad \times \iint |\Psi_1(z)|\, |\Psi_1(z')|\, (1+|z'|)^{\alpha+\frac{D}{2}} (1+|z|+|z'|)^{|\beta|-2-3\alpha}\, dz\, dz'\end{aligned} \tag{4.45}$$

Ψ_1 is a Schwartz function, so the integral is a macroscopic constant. Replacing μ and λ by their value, and because $\sqrt[4]{t} + \sqrt[4]{T-2t} \lesssim \sqrt[4]{T}$, we get :

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |((EF_t)_t - EF_{2t})_{T-2t}(x)|^q)^{\frac{1}{2}} \lesssim (\sqrt[4]{t})^{3\alpha+\frac{D}{2}} (\sqrt[4]{T} + R)^{|\beta|-2-3\alpha} \tag{4.46}$$

Recall that $t = 2^{k-n}T$, for $k = 0, \ldots, n-1$. Plugging in the telescoping sum, we get :

$$\begin{aligned}\Big(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(EF_\tau - F_{x\tau})_{T-\tau}(x)|^q\Big)^{\frac{1}{2}} &\lesssim (\sqrt[4]{T} + R)^{|\beta|-2-3\alpha} \sum_{k=0}^{n-1} (\sqrt[4]{2(k-n)T})^{3\alpha+\frac{D}{2}} \\ &\lesssim (\sqrt[4]{T})^{3\alpha+\frac{D}{2}} (\sqrt[4]{T} + R)^{|\beta|-2-3\alpha} \sum_{i=1}^{n} \rho^i \quad (\rho \coloneqq (\sqrt[4]{2})^{-3\alpha-\frac{D}{2}}) \\ &\lesssim (\sqrt[4]{T})^{3\alpha+\frac{D}{2}} (\sqrt[4]{T} + R)^{|\beta|-2-3\alpha}\end{aligned}$$

where the final inequality is justified by the fact that $\rho < 1$ because of assumption $3\alpha + \frac{D}{2} > 0$.

Now, we need to prove that taking $\tau \to 0$, we get (4.34). By the semigroup property, we have :

$$(EF_\tau - F_{x\tau})_{T-\tau}(x) = (EF_\tau)_{T-\tau}(x) - (\mathrm{d}\Gamma_x^* \Pi_x^-)_{\beta T}(x).$$

Let's prove $(EF_\tau)_{T-\tau}(x) \longrightarrow \delta\Pi_T^-(x)$ for all $x \in B_R$. By definition :

$$(EF_\tau)_{T-\tau}(x) = \int \mathrm{d}y\, \psi_{T-\tau}(y)\, (\mathrm{d}\Gamma_{x-y}^* \Pi_{x-y}^-)_{\beta\tau}(x-y). \tag{4.47}$$

By (4.24) and the pointwise convergence $\psi_{T-\tau} \to \psi_T$, the integrand converges a.s. to $\psi_T(y)\, \delta\Pi_\beta^-(x-y)$ as $\tau \to 0$. To apply the dominated convergence theorem, we claim that $(\mathrm{d}\Gamma_z^* \Pi_z^-)_\beta$ is of polynomial growth in z. Since ψ is a Schwartz function and therefore decays faster than any polynomial, this provides an integrable dominating function. Applying Fatou's lemma successively in the probability space and then in B_R, the uniform bound (4.40) passes to the limit and yields (4.34). □

Remarque 4.1. To see that $(\mathrm{d}\Gamma_z^* \Pi_z^-)_\beta$ has polynomial growth, recall that we work with a fixed $\rho > 0$. So all the objects constructed from ξ_ρ - which is of polynomial growth - by convolutions, and polynomial operations, are of polynomial growth.

A more pedestrian argument consists in relying on induction hypothesis :

$$(\mathrm{d}\Gamma_z^*\Pi_z^-)_\beta = \sum_{\substack{\gamma\neq\mathrm{pp} \\ |\gamma|_\prec<|\beta|_\prec}} (\mathrm{d}\Gamma_z^*)_\beta^\gamma \Pi_{z\gamma}^-.$$

First, we prove a.s. polynomial growth of $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$, $\gamma \neq$pp. By (4.33) with $R = 2^k$:

$$\int_{B_{2^k}} \mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)\right|^q \,\mathrm{d}x \lesssim 2^{k(D+2(|\beta|-|\gamma|))}.$$

Set $Z_k := \left(\int_{B_{2^k}} \left|(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)\right|^q \,\mathrm{d}x\right)^{2/q}$. By Markov's inequality, for any $\varepsilon > 0$:

$$\mathbb{P}\Big(Z_k > 2^{k(D+2(|\beta|-|\gamma|)+\varepsilon)}\Big) \leq \frac{\mathbb{E}[Z_k]}{2^{k(D+2(|\beta|-|\gamma|)+\varepsilon)}} \lesssim 2^{-k\varepsilon}.$$

This is summable in k, so by the Borel-Cantelli lemma, a.s. for all k large enough:

$$\int_{B_{2^k}} \left|(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)\right|^q \,\mathrm{d}x \lesssim_\omega 2^{kN_\varepsilon},$$

for some $N_\varepsilon > 0$. That is, $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$ has a.s. polynomial L^q-growth in x. Because $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$ is smooth (we work at a fixed $\rho > 0$), Sobolev'inequality gives us almost sure polynomial growth uniformly in x.

Secondly by induction hypothesis, (4.26) holds for $|\gamma|_\prec < |\beta|_\prec$.

Chapter 5

Conclusion

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